

06 · Is spend *causing* sales? — causal MMM (pymc-marketing)

July 13, 2026

Runs in the legacy environment (pymc<6): `make env-legacy, kernel cmp-legacy.`

The business decision. Brand search converts like crazy — should we pour budget into it? Be careful: brand search mostly **catches** demand that upper-funnel channels (TV) and seasonality already created, so crediting it for those conversions and shifting budget its way would be a mistake.

The concepts

- **MMM (Marketing Mix Model)** — a regression of sales on each channel’s spend over time (usually weekly), used to attribute sales to channels and set budgets. It’s the “top-down” alternative to click-tracking.
- **Adstock (carry-over)** — advertising doesn’t only work the week it runs; its effect *decays over subsequent weeks*. Adstock transforms raw spend into an “effective” exposure that carries over.
- **Saturation (diminishing returns)** — the 10th €1,000 into a channel buys less lift than the 1st. A saturation curve bends over as spend rises; its *slope* is the **marginal** return that should drive the next euro of budget.
- **Last-click attribution** — crediting the conversion to the final touch (often brand search). It systematically **over-credits** demand-capture channels, which is the error a causal MMM aims to fix.
- **Incremental vs mediated** — the *incremental* contribution of a channel is what it *caused*; a channel can show a big correlation with sales while causing very little (it was downstream of the real driver).

The arc of this notebook — a cautionary tale with a resolution

A plain MMM regresses sales on all channels and is fooled by a **confounder** like seasonality (it lifts sales *and* drives brand-search spend). The textbook fix is a **causal MMM**: hand it a **DAG** and let the backdoor logic (nb 05) control for the confounder. This notebook shows that fix is **necessary but not sufficient**, in three acts — preceded by a fourth thing worth knowing:

0. **Step 0 · the classical read.** Before any Bayesian machinery we fit the MMM the way most agencies actually do: grid-search the adstock decay and the saturation half-point to minimise squared error, then run OLS on the transformed spend with honest (HAC) standard errors. It produces a clean answer — and we show that the data barely constrain those transform constants at all, so the neat interval it reports is **conditional on a choice made in a spreadsheet**. That precise failure is the reason the rest of this notebook exists; §5x returns to grade the Bayesian answer against it, honestly.

1. **The confounding correction works — in a linear model.** A naive OLS attribution that omits seasonality massively over-credits brand_search; adding seasonality as a control collapses that credit toward the small truth. That’s the cleanest way to *see* the confounding, and it’s robust.
2. **The out-of-the-box causal MMM still inverts the ranking.** Fit with library-default priors, the MMM’s *flexible saturation curve* for brand_search **re-absorbs** the seasonal demand the linear control was meant to remove — so it credits brand_search *more* than TV, the opposite of the truth. A DAG plus a linear control does **not** de-confound a channel whose spend is collinear with the confounder.
3. **Experiment-calibrated priors recover it.** Feed in what a small **geo experiment** teaches — that brand_search’s incremental effect is small — as an informative prior on its saturation, and the MMM ranks the channels correctly again. **This is why MMMs must be calibrated with experiments** (Anchor B, nb 07): observational data alone cannot pin the decomposition.

On real data. Swap in your **own weekly spend-by-channel + sales** table plus known confounders (seasonality, price, distribution). Public example: the datasets shipped with Meta’s *Robyn* or *pymc-marketing*’s own tutorials. Always pair an MMM with occasional geo experiments to anchor the levels.

It follows the repo’s fixed **7-step contract**: question → simulate → identify → estimate → validate → decide in € → caveats.

2 · Simulate a ground truth

Weekly sales from a faithful **adstock + Michaelis-Menten saturation** process. **TV** is a big genuine driver. **seasonality** is a **confounder** — it lifts sales directly *and* drives brand-search spend. **brand_search** has only a *small* real effect, but its spend rides the seasonal wave (corr ≈ 0.78), so a naive attribution ignoring seasonality massively over-credits it. Honest answer: TV highly incremental, brand_search only slightly.

The data-generating model — exactly what `dgp.mmm_weekly` implements (defaults & seed in `src/cmp/dgp.py`). Weeks $t = 0, \dots, 155$; seasonality $s_t = \sin(2\pi t/52)$:

$$\begin{aligned}
 x_t^{\text{tv}} &= \max\{1, 40 + 8 s_t + \varepsilon_t^{\text{tv}}\}, \quad \varepsilon_t^{\text{tv}} \sim \mathcal{N}(0, 8^2); & x_t^{\text{bs}} &= \max\{1, 25 + 11 s_t + \varepsilon_t^{\text{bs}}\}, \quad \varepsilon_t^{\text{bs}} \sim \mathcal{N}(0, 7^2) \\
 a_t(x; \lambda) &= x_t + \lambda a_{t-1}(x; \lambda) & & \text{(geometric adstock / carry-over)} \\
 f(a; \alpha, \kappa) &= \frac{\alpha a}{\kappa + a} & & \text{(Michaelis–Menten saturation)} \\
 \text{sales}_t &= 100 + \underbrace{f(a_t(x^{\text{tv}}; 0.4); 60, 50)}_{\text{TV: big, real}} + \underbrace{f(a_t(x^{\text{bs}}; 0.2); 8, 40)}_{\text{brand search: small, real}} + \underbrace{35 s_t}_{\text{season} \rightarrow \text{sales}} + \varepsilon_t, & & \varepsilon_t \sim \mathcal{N}(0, 4^2).
 \end{aligned}$$

The confounding is the **two appearances of s_t** : it drives brand-search spend (the $11 s_t$ term — people search the brand more in peak season; corr ≈ 0.78) *and* sales directly (the $35 s_t$ term). An MMM that omits seasonality hands much of that $35 s_t$ to brand search. Note the generator uses the MMM’s own functional form (adstock → saturation), so the model is well-specified — any remaining failure is about *identification*, not fit. **Identification** is whether the data can, even in principle, pin down the answer uniquely: a model can fit the sales series perfectly and still leave

the *split* of credit between two channels undetermined — that gap between ‘fits’ and ‘is identified’ is the whole subject of this notebook.

TRUE incremental sales - TV 5,263 brand_search 518 (small)

	date_week	tv	brand_search	seasonality	sales
0	2023-01-02	42.870187	35.579322	0.000000	134.900978
1	2023-01-09	53.049712	17.732754	0.120537	137.863335
2	2023-01-16	27.623875	23.486826	0.239316	142.263680
3	2023-01-23	56.329747	34.885672	0.354605	159.788875
4	2023-01-30	43.339248	33.736743	0.464723	152.485792

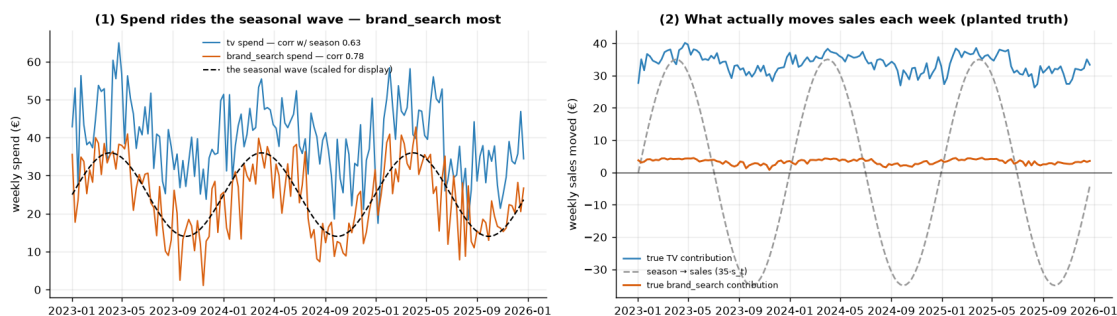
First — look at the data: the confounding, made visible

Two pictures to hold in your head for the rest of the notebook, drawn **before any model sees the data**:

1. **Spend rides the seasonal wave.** brand_search’s spend is *built* from the same seasonal index that lifts sales, so it tracks the dashed wave far more tightly than TV does (both correlations are printed in the legend). Whatever the season does to sales will look, to a model that ignores the season, like brand_search doing it.
2. **Most of what brand_search appears to move is the season.** The right panel plots the *planted* weekly components of sales: TV’s contribution is large, the ± 35 seasonal swing is larger still — and brand_search’s true contribution is the near-flat line hugging the bottom. An attribution method must hand the wave back to the season; everything that follows is about whether it does.

corr(spend, seasonality): brand_search 0.78 vs tv 0.63 - the confounder is entangled

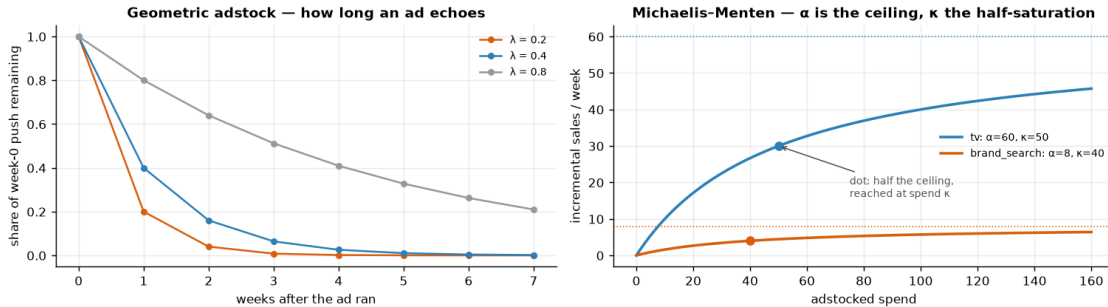
with brand_search; true weekly contributions: TV ~ €34, brand_search ~ €3, season swing +/-€35.



The two transforms, before we use them in anger

The MMM’s channel response is built from the two primitives the generator above already used — worth ten seconds of intuition each before a sampler spends minutes on them. **Adstock** answers “how long does an ad echo?”: with geometric decay λ , a euro of spend still exerts λ^ℓ of its initial push

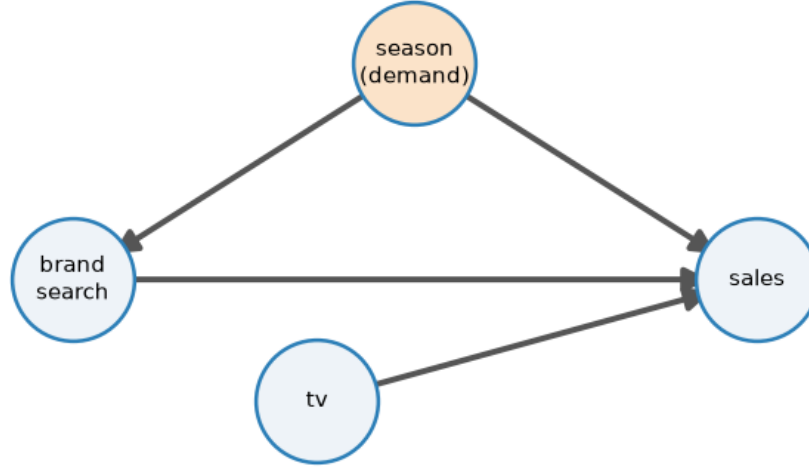
ℓ weeks later, so $\lambda = 0.2$ is a one-week echo while $\lambda = 0.8$ still echoes seven weeks out. **Saturation** answers “*what is the next euro worth?*”: the Michaelis–Menten curve $f(a) = \alpha a / (\kappa + a)$ climbs toward a **ceiling** α — the most the channel could ever add per week — and passes half of it exactly at $a = \kappa$, the **half-saturation** point. Keep the right panel in mind for Step 4b: the experiment-calibrated prior acts on the *ceiling* α , the parameter that says how much a channel could possibly deliver — precisely the number a lift test measures.



3 · Identify — the marketing DAG picks what to control for

DAG: `seasonality` $\rightarrow$ `{sales, brand_search}`, `tv` $\rightarrow$ `sales`, `brand_search` $\rightarrow$ `sales`. The backdoor criterion says **seasonality is a confounder** $\rightarrow$ **control for it**; omitting it is exactly the last-click mistake that over-credits brand search. (This DAG has a confounder but no **mediator** — a variable sitting *on* the causal path from spend to sales, which you would deliberately *not* control for; nb 05 covers that distinction.) The **estimand** — the precise quantity we want to estimate — for channel c is its **incremental contribution** — sum over weeks of $\mathbb{E}[\text{sales} \mid \text{do}(\text{spend}_c = \text{observed})] - \mathbb{E}[\text{sales} \mid \text{do}(\text{spend}_c = 0)]$, holding the other channels at their observed spend (how much of sales would vanish if we switched channel c off). Here $\text{do}(\cdot)$ denotes an **intervention** — setting spend by decree rather than merely observing it (nb 05). **ROAS** (return on ad spend) = incremental contribution $\div$ spend, each with a posterior — note this is *revenue*, not profit, so at $1\times$ you break even only on a 100% margin (we return to margins in Step 6). Identification comes from the backdoor set `{seasonality}`; we hand the DAG to the MMM via `dag`, `treatment_nodes`, `outcome_node`.

Backdoor: seasonality → {brand_search, sales} — control for it



seasonality (orange) is the confounder: brand_search ← seasonality → sales is the backdoor path to block; tv has no backdoor, so its arrow into sales is clean.

Backdoor adjustment set the causal layer will use: ['seasonality']

Step 0 · The classical read (no likelihood, no priors, no sampler)

Before any Bayesian machinery, what does a competent analyst *without* it do here? Exactly what most agencies do, and it is worth doing well, because its **specific failure** is what motivates everything below. The estimand is unchanged from §3 — each channel’s incremental contribution, and its ROAS — and so is the identification argument: the backdoor set is {seasonality}, so seasonality enters the regression as a control. The causal work lives in that DAG, not in the estimator. What changes is only how we *estimate*.

The obstruction. Write the MMM as a regression. Adstock the spend, saturate it, add the control:

$$a_{c,t}(\lambda_c) = x_{c,t} + \lambda_c a_{c,t-1}(\lambda_c), \quad g_{c,t}(\lambda_c, \kappa_c) = \frac{a_{c,t}(\lambda_c)}{\kappa_c + a_{c,t}(\lambda_c)}, \quad \text{sales}_t = \beta_0 + \sum_c \alpha_c g_{c,t}(\lambda_c, \kappa_c) + \gamma s_t + u_t.$$

Note what the algebra just did: dividing the Michaelis–Menten curve $f_c(a) = \alpha_c a / (\kappa_c + a)$ by its ceiling leaves the **unit-ceiling basis** $g_{c,t} \in (0, 1)$, and the ceiling α_c comes back out as the plain OLS **coefficient** on that basis. So *given* the transform parameters (λ_c, κ_c) the model is linear and OLS estimates α_c , γ and β_0 in closed form. The decay λ_c and the half-saturation κ_c , however, enter **non-linearly** — and there is no closed form for them, no OLS interval for them, and (as we are about to see) barely any information about them in a weekly series whose spend rides the seasonal wave.

The classical move: profile them out. Concentrated (profile) least squares — grid-search the four transform parameters, and at each candidate solve the linear part exactly:

$$(\hat{\lambda}, \hat{\kappa}) = \arg \min_{\lambda, \kappa} \underbrace{\min_{\beta_0, \alpha, \gamma} \sum_t \left(\text{sales}_t - \beta_0 - \sum_c \alpha_c g_{c,t}(\lambda_c, \kappa_c) - \gamma s_t \right)^2}_{\text{SSE}(\lambda, \kappa) \text{ — the OLS fit at that transform}},$$

then report OLS at $(\hat{\lambda}, \hat{\kappa})$ **as if those were known**. That last clause is the whole story of this section. It is not a strawman: it is what a careful agency does — and what a careless one does by simply *asserting* “ $\lambda = 0.5$, $\kappa =$ half of average spend, everyone uses those”. Either way the transform is **fixed by fiat**, and the reported interval is conditional on it.

The estimator and its covariance, named. OLS on the transformed regressors, with **Newey–West HAC** standard errors. Weekly sales are serially correlated — adstock alone guarantees it (this week’s sales inherit last week’s exposure), and whatever seasonal structure the single s_t control fails to absorb persists across weeks — so the iid variance formula is simply not the right one here, and an interval built on the wrong standard error is a confident lie. Lag truncation follows the usual rule of thumb, $\lfloor 4(n/100)^{2/9} \rfloor \approx 4$ weeks. The cell prints the residual autocorrelation that motivates the choice *and* the factor by which HAC moved each standard error — and note that “correct” does not automatically mean “wider”: HAC re-weights, and where the seasonal control has already absorbed most of the persistence it can come out **tighter** than iid. The point is never that HAC inflates; it is that the covariance was *checked* rather than defaulted.

Contribution and ROAS then follow from the fitted ceiling by simple arithmetic — the CI carries over by the same scaling, because both are **linear** in $\hat{\alpha}_c$:

$$\widehat{\text{contrib}}_c = \hat{\alpha}_c \sum_t g_{c,t}(\hat{\lambda}_c, \hat{\kappa}_c), \quad \widehat{\text{ROAS}}_c = \frac{\widehat{\text{contrib}}_c}{\sum_t x_{c,t}}.$$

Profiled 41,616 (lambda, kappa) settings by OLS SSE; best $R^2 = 0.9804$.
The winner - these four numbers are now FIXED BY FIAT, no interval attached:

```
tv          lambda = 0.25   kappa = 197.9 (= 5.0x mean weekly spend)
            ^ kappa sits AT THE EDGE of the band: the SSE never
              turned around - no interior optimum exists.
brand_search lambda = 0.00   kappa = 11.6 (= 0.5x mean weekly spend)
planted truth: tv lambda=0.40, kappa=50.0
              brand_search lambda=0.20, kappa=40.0
```

OLS at the profiled transform * Newey-West HAC (maxlags=4) * 90% CIs * n = 156

	channel	ceiling	alpha (€/wk)	contribution (€)	true (€)
tv	77.6	[57.6, 97.6]	2,526	[1,875, 3,178]	5,263
brand_search	5.2	[-1.9, 12.4]	528	[-194, 1,249]	518

	channel	ROAS (x)	true (x)
tv	0.41	[0.30, 0.51]	0.85
brand_search	0.14	[-0.05, 0.33]	0.14

Why HAC: the residuals carry lag-1 autocorrelation +0.18 - the iid variance formula does not apply here.

HAC / iid standard error: x0.96 (tv), x1.13 (brand_search).

It came out TIGHTER than iid: HAC re-weights, it does not only inflate. The point is correctness, not conservatism.

GRADE vs the planted truth

```
ranking : classical says TV > brand_search; truth is TV first -> CORRECT
tv      : contribution 0.48x the truth * 90% ROAS CI holds it? NO
brand_search: contribution 1.02x the truth * 90% ROAS CI holds it? YES
-> the classical fit gets the RANKING right here - where $4a's
    default Bayesian MMM will not. Hold the applause: it bought that
    by FIXING the transform, an assumption it never defended. Run the
    same profiling on the shorter FAST series (CMP_FAST=1) and it
    inverts outright. The next cell measures how little the fit ever
    had to say about it.
```

What we just got, and what it cost. The table prints a complete, professional-looking answer: a ceiling per channel, a contribution, a ROAS, each with a 90% interval computed off standard errors that were *corrected for the serial correlation of weekly sales*. Grade it honestly against the planted truth — the printed lines do, on the ranking and on the levels, and the read-out after the next cell says plainly what they show.

But look again at the four numbers above the table. $\hat{\lambda}$ and $\hat{\kappa}$ came out of a grid search, and they carry **no interval at all**. The regression that produced those neat CIs was handed them as if they were *known constants*. So the honest question is not “is the point estimate good?” but:

How much would the answer move if I had fixed the transform somewhere else — somewhere the data cannot rule out?

That is a question the classical fit is structurally unable to answer, because it *conditioned on the answer*. So we ask it from the outside, two ways: (1) re-fit across a grid of (λ, κ) values a practitioner could defend a priori — the “everyone uses 0.5” spreadsheet choice — and (2) compute, properly, the set of transform values the data genuinely **cannot distinguish**: the profile (concentrated) likelihood region, using the standard Gaussian result that $n \log(\text{SSE}(\lambda, \kappa)/\text{SSE}(\hat{\lambda}, \hat{\kappa}))$ is asymptotically χ_4^2 under the null that (λ, κ) is the truth — the very region whose existence the “fix it and forget it” workflow denies.

The 90% profile region for (λ, κ) holds 25,340 of the 41,616 searched settings - 61% of the grid. Inside it:

```
lambda_tv spans [0.00, 0.55] * lambda_bs [0.00, 0.80] (band: [0, 0.8])
kappa_tv spans [8, 198] * kappa_bs spans [5, 121] €/week
-> the decay and the half-saturation point are, in practice, NOT IDENTIFIED.
```

What that costs the answer:

```
TV ROAS across the region : [0.30, 3.44] (width 3.14)
the REPORTED 90% CI      : [0.30, 0.51] (width 0.21) - 15x narrower
true TV ROAS 0.85x lies OUTSIDE the reported CI, and inside
the transform envelope.
brand_search ROAS across the region: [-2.61, 1.30] (true 0.14x)
ranking TV > brand_search holds in 98% of the region.
(1,374 settings in the region imply a NEGATIVE brand_search
contribution - off the bottom of the right panel. The fit cannot
```

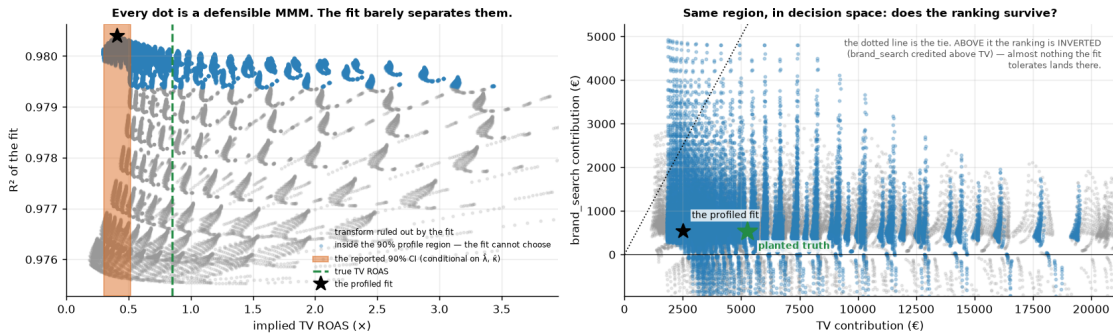
rule those out either.)

"We use industry-standard values" - same data, the transform simply ASSERTED:

lambda (both)	kappa (x mean spend)	R ²	TV ROAS	bs ROAS	ranking
0.2	0.5 0.9800	0.95	0.18	TV first	
0.2	1.0 0.9801	0.64	0.14	TV first	
0.2	2.0 0.9802	0.48	0.11	TV first	
0.4	0.5 0.9797	1.40	0.23	TV first	
0.4	1.0 0.9799	0.89	0.17	TV first	
0.4	2.0 0.9800	0.64	0.13	TV first	
0.6	0.5 0.9779	1.56	0.08	TV first	
0.6	1.0 0.9781	0.98	0.05	TV first	
0.6	2.0 0.9782	0.69	0.03	TV first	

R² spans only [0.9779, 0.9802] across these - indistinguishable fits - while TV's ROAS spans [0.48, 1.56]:
the choice, not the data, is setting the answer.

What this 90% confidence interval does NOT say: that there is a 90% probability the true effect lies inside it. It is a property of the *procedure* - 90% of intervals built this way would cover the truth across repeated samples. This interval either contains the truth or it does not. To get a probability *about the effect itself* - the thing a go/no-go rule like $P(\text{lift} > \text{cost}) \geq 0.9$ actually needs - you need a posterior. That is what the Bayesian section adds.



Read-out — the classical MMM's honest verdict. Take the two panels in turn.

Left. Each dot is one (λ, κ) setting: a complete, defensible MMM, fitted by OLS with the correct control. The vertical axis is how well it fits; the horizontal axis is the TV ROAS it implies. **The cloud is nearly flat** — the fit's R^2 varies in the third or fourth decimal across settings whose implied TV ROAS differ by a factor of several. Blue dots are the ones the data cannot rule out even at the 90% level (the printed profile region typically swallows most of the grid: the decay λ spans almost the entire searched band, and κ likewise). Against that cloud, the orange strip — the interval the classical MMM actually **reports** — is a sliver, and the printed comparison gives the factor. That strip is not wrong arithmetic; it is a correct answer to a **different** question: *given that λ and κ are exactly these values, how precise is the ceiling?* Nobody knows that they are.

Right. The same region in decision space: each dot’s TV contribution against its brand-search contribution, with the dotted line where the two tie. Read the printed share of the region that still ranks TV first. On the committed **FULL** series (156 weeks) that share is high and the profiled fit itself lands on the correct side — the classical MMM gets the **ranking** right, where §4a’s default Bayesian MMM will not. Say it plainly rather than burying it: **out of the box, on this data, the classical estimator beats the Bayesian one on the decision-relevant quantity.** But grade the *reason*, not just the result. It got there by fixing (λ, κ) — an *infinitely strong* assumption, never defended — which happened to forbid the exact manoeuvre that breaks the default MMM. That is luck wearing the clothes of rigour, and the notebook can prove it is luck: run the identical profiling on the shorter **FAST** series and it **inverts the ranking outright**, with no priors and no sampler anywhere in sight. Same estimator, same DGP, opposite budget recommendation. Meanwhile the **levels** wander across a multiple-fold range in either run, and TV’s reported 90% ROAS interval — the one that would be pasted into the deck — can miss the true ROAS entirely (the cell prints whether it did). An interval that excludes the truth is not a small sin; it is the whole sin.

And the spreadsheet table. The “industry standard values” block is the same failure without the grid search: $\lambda = 0.2$ versus $\lambda = 0.6$, κ at half versus twice mean spend — assertions any planner would sign off — move the implied TV ROAS across a wide band while the R^2 column barely twitches. **The data are not choosing. Someone in a meeting is.**

The guardrail (printed above, in the same voice every notebook uses): a confidence interval is a property of the *procedure*, not a probability statement about the effect. And here it is worse than usual — the procedure’s interval is *conditional on transform parameters chosen by fiat*, so it understates total uncertainty by the factor printed above, and it cannot in principle answer the question the budget meeting asks: $P(\text{TV’s ROAS} > \text{brand search’s ROAS})$. What the classical fit is missing is not arithmetic. It is (i) a way to carry the uncertainty in λ and κ into the answer instead of conditioning it away, and (ii) a way to let a **geo experiment** speak to the one quantity the observational data is silent about. Both of those are what §4 buys — and §5x will audit, honestly, how much of it Bayes actually delivers.

4 · Estimate — fit the causal MMM twice

We fit the **same DAG-aware MMM twice**, to make the lesson unmissable:

- (a) **default priors** — the honest, out-of-the-box causal MMM. Watch it *invert* the ranking.
- (b) **experiment-calibrated priors** — encode what a geo experiment teaches (brand_search’s incremental effect is small) as an informative prior on its saturation ceiling, and watch the ranking snap back.

The fitted model, in symbols. Per channel $c \in \{\text{tv}, \text{brand_search}\}$, spend is adstocked over an 8-week window, saturated, and summed with the control:

$$\text{sales}_t \sim \mathcal{N}\left(\beta_0 + \sum_c f_c(a_c(x^c)_t) + \gamma s_t, \sigma^2\right), \quad a_c(x)_t \propto \sum_{\ell=0}^7 \lambda_c^\ell x_{t-\ell}, \quad f_c(a) = \frac{\alpha_c a}{\kappa_c + a},$$

geometric adstock (`GeometricAdstock(l_max=8)`) and Michaelis–Menten saturation, with each channel’s decay λ_c , ceiling α_c and half-saturation κ_c **estimated** from the data. Fit (a) leaves their priors at library defaults; fit (b) replaces the prior on brand-search’s ceiling α_{bs} with the experiment-informed one. Note the functional form matches step 2’s generator exactly — this

is the *well-specified best case*, which is what makes the ranking inversion in (a) a lesson about identification rather than model misfit. (pymc-marketing scales variables internally; the equations are written in original units for readability.)

The priors, made explicit. “Library defaults” in fit (a) means, per channel c (all on **max-abs-scaled** spend and sales — pymc-marketing divides each channel’s spend and the sales series by its own maximum before fitting; Step 4b returns to why that matters):

$$\begin{aligned}\lambda_c &\sim \text{Beta}(1, 3), & \alpha_c &\sim \text{Gamma}(\mu = 2, \sigma = 1), & \kappa_c &\sim \text{HalfNormal}(1), \\ \beta_0, \gamma &\sim \mathcal{N}(0, 2), & \sigma_y &\sim \text{HalfNormal}(2).\end{aligned}$$

The one to stare at is the saturation **ceiling** α_c , in scaled-sales units: a prior mean of 2 says “this channel could plausibly add up to about *twice the all-time weekly sales peak*, every week” — weakly-informative in the worst sense, generous exactly along the direction the data cannot push back on. Fit (b) replaces only brand-search’s copy of that one prior.

Convergence housekeeping: this is the hardest posterior geometry in the cookbook — the FULL profile needs `target_accept = 0.99` and deeper trees to reach $\hat{R} \approx 1.01$ (the diagnostics — $\hat{R}$, ESS, divergences — are defined in notebook 01’s “did the sampler converge?” box). Even then, expect fit (a) to log **a dozen-odd** divergences and fit (b) to sit exactly at the 1.01 edge — and in FAST mode either fit’s short chains can disagree outright ($\hat{R} \gg 1.01$). Read those complaints as **information, not nuisance**: the sampler is struggling along the very season/brand-search ridge — a flat valley of near-equivalent answers — that breaks identification (we show it in the flesh right after the fit). We quote fit (a) only for its — robustly wrong — ranking, never for its intervals.

A caution that runs through both: **MMMs are miscalibrated in absolute terms** — with channel spend riding seasonal demand, a short weekly series can’t fully separate “channel drove sales” from “season drove both”, so fitted contributions run high. What we’re after is the **ranking** and each channel’s **ROI verdict**.

```
(a) default-prior MMM convergence: max r-hat 1.010 - min ESS 381 - divergences
14
(a) default contribution - TV 10,390   brand_search 36,007
      true: TV 5,263, brand_search 518  -> the MMM ranks brand_search ABOVE TV
(inverted!).
```

Why the out-of-the-box MMM inverts the ranking

The default MMM credits `brand_search` *above* TV — backwards. The mechanism is worth seeing, because it’s a real trap:

- `brand_search`’s spend is *built* to ride the seasonal demand wave (Step 2 constructs it from the same seasonal index that drives sales), and the fit lets its flexible saturation curve quietly **re-absorb the seasonal demand** the linear **seasonality** control was meant to remove.
- Because `brand_search` rides the seasonal wave more tightly than TV does, it wins that tug-of-war and books a huge contribution (tens of thousands vs a true ~500), while a large **negative baseline** offsets it — the classic broken **level decomposition** — how the model splits the *absolute* level of sales across baseline + channels — of a non-identified MMM.

- The DAG was right and the control was present; they still weren't enough. **A linear confounder-control does not de-confound a channel whose spend is collinear**** with (moves almost in lockstep with) the confounder and enters through a flexible functional form.** Observational data alone can't break the tie.

Those are three checkable claims — collinear spend, a posterior tug-of-war, a negative baseline — so the next cell checks them, in the data and in fit (a)'s own posterior.

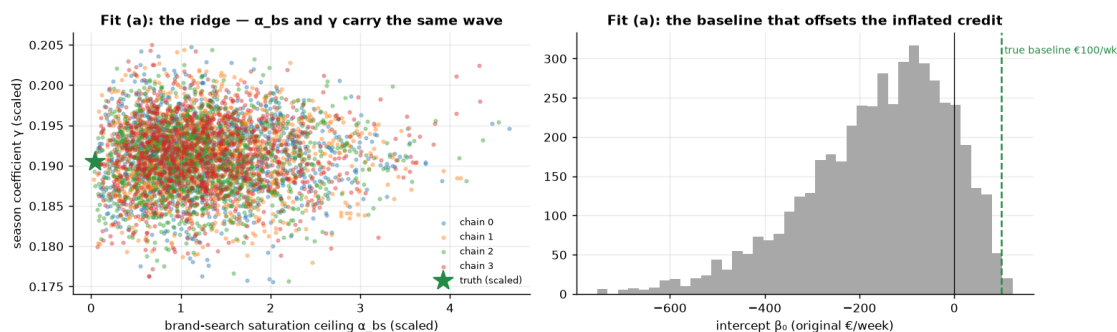
spend-confounder correlations (the tv row is the contrast):

	tv	brand_search	seasonality
tv	1.00	0.54	0.63
brand_search	0.54	1.00	0.78
seasonality	0.63	0.78	1.00

draw-level `corr(alpha_bs, gamma) = +0.01` * per-chain `alpha_bs` posterior means: [1.27 1.3 1.25 1.24]

truth in scaled units: `alpha_bs ~ 0.044`, `gamma ~ 0.191` - the green star, far from the mass.

intercept (original scale): mean €-160/week vs planted baseline €100/week -> **NEGATIVE**: the level decomposition is broken.



Read-out. *Left*: every dot is a posterior draw of (brand-search ceiling α_{bs} , season coefficient γ). The truth (green star) has a tiny ceiling; the posterior instead spreads along a **ridge** of near-equivalent decompositions — draws (and, at FAST settings, whole chains) trade the seasonal wave back and forth between the two parameters, because the likelihood is near-flat along that direction. *Right*: to pay for the inflated brand-search credit, the intercept is dragged far below the planted €100/week — the classic broken level decomposition of a non-identified MMM, exactly the “large negative baseline” claimed above. When two regressors carry the same wave, the posterior does not pick one: it spreads over every split the data cannot distinguish, and under the default priors the flexible saturation curve books the credit. **No amount of extra sampling fixes this — only information from outside the dataset can.** Which is the move of the next section.

The fix — priors calibrated from a geo experiment

The standard remedy is exactly the discipline of Anchor B (the cookbook's ‘calibrate MMMs with experiments’ rule, nb 07): run a **small geo experiment** — which randomizes a channel's spend across matched groups of regions (some markets get the ads, some don't) and reads the sales gap

as that channel’s *true* incremental lift, uncontaminated by the seasonality the observational data can’t net out (full treatment in nb 07) — to learn that brand_search’s *incremental* effect is small, and feed that back as an **informative prior on its saturation ceiling** (saturation_alpha). TV, whose incremental role we haven’t independently pinned down, keeps a weakly-informative prior. This is not putting a thumb on the scale — it’s using experimental evidence exactly where observational data is silent.

From experimental euros to a prior — the mapping, explicitly. The geo test (nb 07) delivers an estimate of brand-search’s total incremental sales over its test window, $\hat{L} \pm \text{SE}$. The MMM says brand-search’s weekly contribution at the observed operating point is $f(\bar{a}) = \alpha_{bs} \bar{a} / (\kappa_{bs} + \bar{a})$, with $\bar{a}$ the typical adstocked spend. Matching the model to the experiment therefore pins the ceiling:

$$\alpha_{bs} \approx \underbrace{\frac{\hat{L}}{n_{\text{weeks}}}}_{\text{weekly lift the test measured}} \cdot \underbrace{\frac{\kappa_{bs} + \bar{a}}{\bar{a}}}_{\text{ceiling from the operating point}}, \quad \alpha_{bs}^{(s)} \sim \text{Gamma}(\mu = 0.3, \sigma = 0.15),$$

with σ scaled from the experiment’s SE — wide enough that the MMM’s data may still disagree, tight enough to break the ridge. One unit convention to keep straight: pymc-marketing fits on **max-abs-scaled** spend and sales, so the prior acts on $\alpha_{bs}^{(s)} = \alpha_{bs} / \max_t(\text{sales}_t)$ — which is why the 0.3 is not comparable to the DGP’s $\alpha_{bs} = 8$ (sales units) without that conversion. A mean of 0.3 says “the test showed brand search can move at most a modest fraction of peak weekly sales”; TV keeps the default $\text{Gamma}(\mu = 2, \sigma = 1)$ — an implied ceiling near *twice* peak weekly sales, i.e. essentially unconstrained, honest about the fact that no experiment has pinned TV down. Step 5’s sensitivity sweep then shows the conclusion survives anywhere in the experiment-plausible range — the answer to “didn’t you just pick the prior you wanted?”.

Production note. pymc-marketing can also ingest the experiment *directly*: `MMM.add_lift_test_measurements(df_lift)` appends each lift measurement to the likelihood as a distance penalty on the saturation curve’s rise over the test’s spend delta — a cleaner, multi-test-ready alternative to hand-translating one experiment into one Gamma prior, and the recommended pattern outside the classroom.

```
(b) calibrated-prior MMM convergence: max r-hat 1.010 - min ESS 376 -
divergences 0
(b) calibrated contribution - TV 12,820    brand_search 2,262
    true: TV 5,263, brand_search 518  ->  ranking RESTORED: TV above
brand_search.
```

5 · Validate — fit vs identification, the correction, recovery, and robustness

Six views, from “does it fit?” to “does the conclusion survive?”:

1. **Posterior predictive check, both fits** — and the deliberately uncomfortable punchline that *fit quality cannot arbitrate between the inverted and the calibrated decomposition*.
2. **The confounding correction, several ways** — naive OLS (omit season) over-credits brand_search; OLS + **season** collapses it toward truth; the **default** causal MMM *inverts*

it; the **calibrated** MMM restores the right ranking. The contrast between the two MMMs is the headline.

3. **ROAS with uncertainty (calibrated MMM)** — each channel's posterior ROAS vs the break-even line. $\text{ROAS} = \text{contribution} \div (\text{fixed observed spend})$, so it carries the **same relative miscalibration as the contribution** — dividing by a constant spend cannot remove the bias. What is robust is the ROAS **ranking** (and brand_search's below-break-even verdict), **not** the absolute ROAS *levels*, which are biased by the same factor as the contributions.
4. **Saturation curves** — contribution vs spend per channel, showing diminishing returns (the shape that sets *marginal* ROI and hence where the next euro should go).
5. **Parameter recovery vs the planted truth** — grade the *parameters*, not just the contributions, to see exactly where the non-identification lives (and why the calibrated prior is surgical).
6. **Prior sensitivity** — the fix as a robustness *range*, not a single hand-picked point.

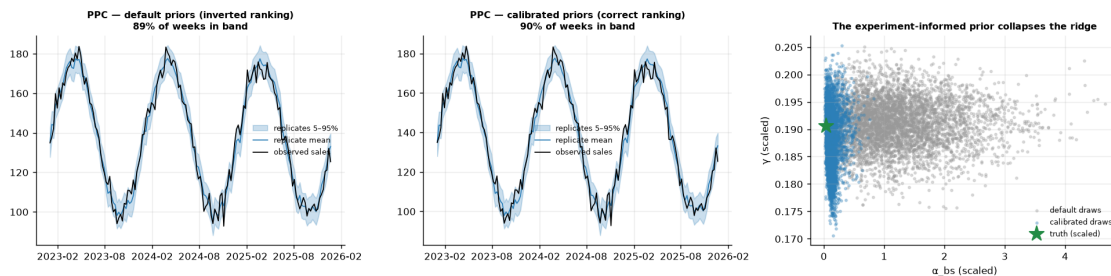
Then **5x** closes the section by putting the three estimators — Step 0's classical MMM, the default Bayesian MMM and the calibrated one — side by side on the same estimand, and asking what the posterior actually bought (and what it did not).

View 1 — the posterior predictive check that cannot save you

Before reading any decomposition off either fit, we ask the standard model-criticism question (nb 01, Step 4): *can the model reproduce the data it was trained on?* For each fit we draw replicated sales series from the posterior predictive, lay the replicates' 5–95% band over the observed series, and print the share of observed weeks inside the band ($\approx 90\%$ means calibrated). Watch for what does **not** happen: the model with the inverted, wildly wrong decomposition does not fit any worse.

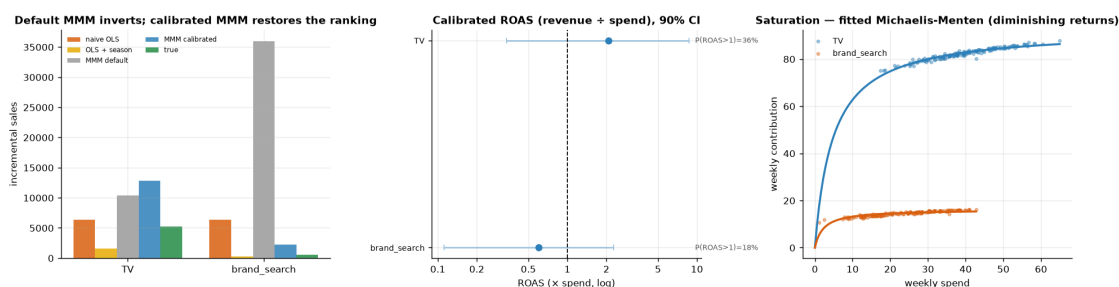
share of observed weeks inside the 5-95% replicate band - default 89% *
 calibrated 90% (nominal ~ 90%)
 replicate-mean RMSE (€/week) - default 3.9 *
 calibrated 3.9 (DGP noise sigma = 4)

Two decompositions that disagree by an order of magnitude on brand_search reproduce the observed sales series near-identically - the likelihood cannot tell them apart. The ridge is where they hide.



Read-out. The two left panels are, for practical purposes, the same picture: near-identical bands, coverage and RMSE (printed above) — yet the left model credits `brand_search` with tens of times its true contribution. This is the precise sense in which the notebook’s failure is **identification, not fit**: the generator uses the MMM’s own functional form (Step 2’s “well-specified” remark), so both fits sit near the same likelihood value, and no amount of posterior-predictive checking can distinguish the inverted decomposition from the correct one. Keep running PPCs — a model that *cannot* reproduce the series is disqualified before you read anything off it — but treat them as **necessary, never sufficient** for a causal claim. *Right panel:* what the experiment-informed prior actually did. The default posterior (grey) spreads along the ridge; the calibrated posterior (blue) is pinned at the low-ceiling end, beside the truth. The prior did not overrule the data — it **selected among answers the data could not rank**.

```
brand_search credit - naive 6,401 | OLS+season 222 | MMM default 36,007 | MMM
calibrated 2,262 | true 518
calibrated ROAS - TV 2.08x [90% 0.34,8.67] * brand_search 0.60x [90%
0.11,2.28]
TRUE ROAS          - TV 0.85x * brand_search 0.14x (both below 1x: the MMM
recovers the ranking but overstates levels)
```



How to read this. *Left* tells the whole story in one panel. **Naive OLS** (orange) over-credits `brand_search`; **OLS + season** (gold) collapses that credit toward the small truth — the confounding correction, working. Then the two MMMs: the **default** causal MMM (grey, the failure mode) *inverts* the ranking, crediting `brand_search` far above TV — the seasonal-demand-stealing trap — while the **calibrated** MMM (blue) restores TV above `brand_search`, landing beside the **true** bars (green). *Middle* — the decision-relevant **calibrated ROAS**: TV’s mean clears break-even but its 90% interval is wide and dips below 1×, while `brand_search`’s mean sits **below break-even**. So TV is the better channel, but its above-break-even verdict is itself uncertain. And the ranking is only *probably* right: draw-by-draw, TV’s ROAS beats `brand_search`’s about **78%** of the time in the committed FULL run (the $P(\text{TV ROAS} > \text{brand_search ROAS})$ computed in Step 6) — more likely than not, but **short of the 0.8 bar** we’d want before moving real budget. *Right* — the **saturation curves**: contribution flattens as spend rises (diminishing returns), so the *next* euro into a channel is worth its curve’s *slope*, not its average. The honest caveat (Step 7): even the calibrated ROAS *levels* are approximate — anchor them with the same kind of geo experiment that supplied the prior.

View 5 — grade the parameters, not just the contributions

Step 2 planted $\lambda = (0.4, 0.2)$, $\alpha = (60, 8)$, $\kappa = (50, 40)$, a baseline of 100 and a season coefficient of 35 — and the MMM estimates those same quantities, so we can grade them directly (the cookbook’s validation-first premise). One bookkeeping step first: the fitted parameters live in max-abs-scaled units behind a *normalized* adstock, so we map each posterior draw back to original units before comparing — the ceiling as $\hat{\alpha}_c = \alpha_c^{(s)} \cdot \max_t(\text{sales}_t)$, the half-saturation as $\hat{\kappa}_c = \kappa_c^{(s)} \cdot \max_t(x_{c,t}) \cdot S(\hat{\lambda}_c)$ where $S(\lambda) = \sum_{\ell=0}^7 \lambda^\ell$ is the adstock normalizer, and the decay λ_c compares as-is (it is scale-free). The question the table answers is not “are the numbers perfect?” (they will not be — Step 4’s caution about levels applies to parameters too) but **where the error concentrates**.

Parameter recovery per fit - cells show posterior mean [90% CI];
season coeff is EUR/wk per unit s_t.

channel	parameter	true		default		calibrated
tv	adstock decay	0.4	0.234	[0.0473, 0.416]	0.223	[0.0416, 0.411]
tv	ceiling (€/wk)	60.0		103 [43.9, 294]		114 [42.6, 353]
tv	half-sat (€)	50.0		92.4 [1.72, 232]		81.3 [1.29, 222]
brand_search	adstock decay	0.2	0.351	[0.0645, 0.672]	0.336	[0.0388, 0.672]
brand_search	ceiling (€/wk)	8.0		232 [58.2, 478]		23.2 [7.9, 57]
brand_search	half-sat (€)	40.0	0.91	[0.0116, 1.37]		60.8 [0.39, 178]
	- baseline (€/wk)	100.0		-160 [-450, 50.8]		41 [-222, 119]
	- season (€/wk)	35.0		35.1 [33.7, 36.4]		34.5 [33, 36]

Where the error concentrates: brand_search’s fitted ceiling is ~29x its true value under default priors, and ~3x under the calibrated prior - the calibration acts on exactly the one parameter the data cannot pin down. TV’s curve parameters stay loosely identified in BOTH fits (read the CIs above) - which is why Step 6 trusts the ranking, never the levels.

Read-out. The table closes the loop on Step 4b’s “not a thumb on the scale” claim. The default fit’s damage is *concentrated*: brand-search’s ceiling — the parameter that can impersonate the seasonal wave — inflates by an order of magnitude or more, and the level account is dragged along to offset the inflated credit — read the baseline row, far from the planted 100. The calibrated prior moves **that one number** back toward the truth and leaves every other parameter to the data — surgical, because the experiment spoke to exactly the quantity the likelihood is silent about. Note also what calibration does *not* fix: TV’s ceiling and half-saturation remain wide and level-shifted in both fits. Rankings, not levels — the caution of Step 4 holds at the parameter level too.

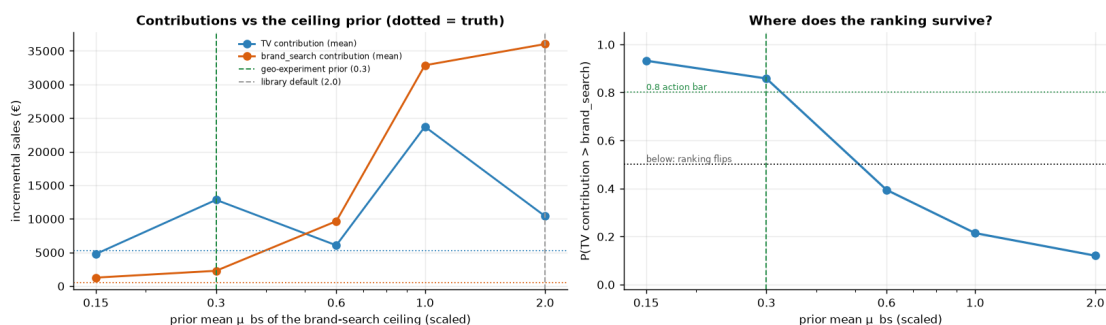
View 6 — “didn’t you just choose the prior that gives the answer you wanted?”

The skeptic’s question, and it deserves a computed answer rather than a shrug. We refit the MMM across a grid of prior means for brand-search’s saturation ceiling — from half our calibrated value up to the library default — keeping the same Gamma shape family throughout ($\mu_{bs} \in \{0.15, 0.3, 0.6, 1.0, 2.0\}$ in scaled units, $\sigma = \mu/2$). The 0.3 and 2.0 grid points *are* the two headline fits, reused rather than refit; the three new fits run at **ranking-grade sampler settings** (short chains — stable posterior means and ranking probabilities, not interval-grade diagnostics, exactly like notebook 01’s stability sweep), so this cell stays cheap in both FAST and FULL profiles. Two curves to read: each channel’s estimated contribution as the prior loosens, and $P(\text{contrib}_{tv} >$

contrib_{bs}) against the 0.5 line (ranking flips) and the 0.8 action bar.

Ranking verdict across the sweep: correct ($P > 0.5$) for $\mu_{bs} \leq 0.3$; flipped by $\mu_{bs} = 0.6$.

The geo experiment would have to be wrong by $\sim 2x$ before the conclusion moved. (Sweep fits are ranking-grade short chains: read the means and the flip point, not intervals.)



Read-out. The conclusion is not an artifact of one hand-set prior. Across the experiment-plausible range of prior means the calibrated ranking — TV above brand_search — holds, and it only gives way as the prior mean approaches the library default, i.e. as we progressively throw the experiment’s information away (the printed line locates the flip for this run). The left panel is the honest dial: loosening the prior slides brand-search’s contribution smoothly from near its dotted truth line back up toward the ridge’s answer, with the data pushing back only weakly anywhere along the way. That is what “the experiment carries the identification” looks like — and it also quantifies robustness: a real geo test would have to be wrong by *about twofold* (roughly double), not a few percent, before this budget decision changed direction.

5x · Point estimate vs posterior — what the Bayesian layer actually bought

Three estimators are now on the table, all targeting the **same estimand** under the **same identification argument** (backdoor-adjust for seasonality): the classical profiled MMM of Step 0, the default-prior Bayesian MMM, and the experiment-calibrated Bayesian MMM. Line them up on the same quantities — contribution, ROAS, the ranking — and on the thing that actually differs: what each is entitled to *say*.

Same estimand, same DAG, three estimators - mean [90% interval]:

estimator	TV contribution (€)	brand_search (€)	ranking
truth	5,263	518	TV first
classical (Step 0)	2,526 [1,875, 3,178]	528 [-194, 1,249]	TV first
MMM default	10,390 [2,113, 44,033]	36,007 [8,803, 74,287]	INVERTED

MMM calibrated 12,820 [2,096, 53,524] 2,262 [423, 8,596] TV first

estimator	TV ROAS (x)	brand_search ROAS (x)
truth	0.85	0.14
classical (Step 0)	0.41 [0.30, 0.51]	0.14 [-0.05, 0.33]
MMM default	1.68 [0.34, 7.13]	9.54 [2.33, 19.68]
MMM calibrated	2.08 [0.34, 8.67]	0.60 [0.11, 2.28]

The interval on TV's ROAS (true = 0.85x) - width, coverage, and what each one conditions on:

classical CI (transform fixed): [0.30, 0.51] width 0.21 -> covers truth? NO
+ the transform envelope : [0.30, 3.44] width 3.14 -> covers truth? YES
(Step 0's profile region - the ignorance the CI above hides)
calibrated posterior CrI : [0.34, 8.67] width 8.33 -> covers truth? YES
(lambda and kappa integrated out, not fixed by fiat)

The decision quantity itself:

posterior $P(\text{TV ROAS} > \text{brand_search ROAS}) = 0.78$ (vs the 0.8 action bar)
The classical arm has no such number. Its nearest analogue - the share of the lambda,kappa settings the fit cannot rule out that rank TV first - is 98%. But that is a statement about a GRID, not a probability about the channels: it cannot be compared to the 0.8 bar.

The honest verdict, in three beats.

1 · Do they agree? Read the ranking row. The *calibrated* Bayesian MMM ranks TV first, as the truth does; the *default* Bayesian MMM inverts it; and the classical fit's verdict depends on which run you are reading — on the committed FULL series it ranks TV first (out-scoring the default MMM, which is the uncomfortable part, and we said so in Step 0), while on the shorter FAST series the same estimator inverts. Both of those classical outcomes were produced by *fixing* (λ, κ) by fiat: an infinitely strong constraint that in one sample forbade the seasonal-wave theft and in the other invited it. That is not robustness; it is a coin landing the right way up, and Step 0's profile region — which contains settings that invert the ranking — is the coin. On *levels* nobody covers themselves in glory: the classical fit and the calibrated MMM miss the true contributions in **opposite directions**, by multiples (read the table). Neither is a euro-grade level estimate, exactly as §6 and §7 insist.

2 · What the posterior gives you that the point estimate cannot. Three things, all visible above. (i) *Uncertainty that includes the transform.* The Bayesian MMM puts a **prior on λ and κ and integrates over them**, so its interval propagates precisely the ignorance the classical CI conditions away. Compare the three intervals printed on TV's ROAS: the classical CI is the narrow one — and it is narrow because it *assumed away* the dominant source of uncertainty; widen it by hand with Step 0's profile region and you land in the same neighbourhood as the posterior. The posterior did not “add” uncertainty, it declined to hide it. Which of them contains the truth is printed above; the narrow one has no monopoly on being right, and every reason to be over-confident. (ii) *A native channel for experimental evidence.* A geo lift test enters the Bayesian model as a **prior on the saturation ceiling** — the exact parameter the test measures (Step 4b) — or, in production, straight into the likelihood via **add_lift_test_measurements**. The classical analogue is a hard constraint or a ridge penalty on α_{bs} : a prior in disguise, with no honest interval

and no principled way to weigh the test’s own standard error against the data. (iii) *The number the budget meeting needs.* $P(\text{TV ROAS} > \text{brand search ROAS})$ is a posterior probability. It is printed above and compared to the 0.8 action bar; the classical arm has no such quantity, only a point ranking, which is why §6’s whole euro decision — the shift-size sweep *with its band*, the value of the experiment — is a Bayesian construction end to end.

3 · What Bayes did NOT buy. It did **not** rescue a non-identified model. That is this notebook’s spine and Step 0 sharpens it into a needle: the flat SSE surface of Step 0 and the posterior ridge of §4a are *the same object seen twice* — one likelihood, near-flat along a direction that trades brand search’s ceiling against the seasonal coefficient. Being Bayesian about it changes the failure’s shape, not its existence, and with the library’s default priors it made it **worse** (the inverted ranking) — a generous $\text{Gamma}(\mu = 2)$ ceiling prior is an active invitation to steal the wave, not a neutral abstention. Priors help only when they carry **real information**. Here exactly one of them did: the geo experiment’s prior on α_{bs} — and View 6 showed the conclusion tracks that prior’s strength, which is the honest confession that **the experiment, not the estimator, is doing the identifying**. Bayes is what let the experiment *in*, coherently, with an interval on the other side. It is not what made the model identified.

6 · Decide, in euros — should we reallocate budget yet?

Budget chases **incremental return**, not last-click credit. One naming point first: what the MMM gives us is **ROAS** — incremental *revenue* per euro of spend — not profit ROI. Break-even at $1\times$ therefore assumes a 100% margin; at a realistic margin m the break-even ROAS is $1/m$ (a 30% margin $\rightarrow 3.3\times$), which we show below.

Now that experiment-calibrated priors have made the *ranking* trustworthy, the point estimates put TV above brand_search, so the *direction* of any reallocation is toward TV. But a direction is not a decision, and the **absolute levels are still miscalibrated**. So we quantify three things: (1) the sales impact of shifting a slice of budget — swept across shift sizes and made **saturation-aware**, so marginal ROAS falls as we pour more into TV; (2) the posterior probability the move actually helps; and (3) the honest gap between the MMM’s ROAS and the *true* ROAS. That last check is the sobering one: it is why, even with the ranking fixed, the absolute “is TV worth it?” question still belongs to a geo test (Anchor B, nb 07), not a still-approximate MMM.

Average vs marginal, in symbols. Two different euros-per-euro numbers get called “ROAS”, and budget moves must be priced with the second one:

$$\text{ROAS}_c = \frac{\sum_t f_c(a_{c,t})}{\sum_t x_{c,t}}, \quad \text{mROAS}_c(\bar{x}) = \left. \frac{\partial f_c}{\partial x} \right|_{x=\bar{x}} = \frac{\alpha_c \kappa_c}{(\kappa_c + \bar{x})^2},$$

the **average** return on every euro already spent versus the **marginal** return on the *next* euro at the current operating point $\bar{x}$ — and on a saturating curve the marginal is always below the average (the curve’s slope keeps falling). The optimal budget split is reached exactly when marginal returns equalize across channels, $\text{mROAS}_{tv} = \text{mROAS}_{bs}$: that equal-marginal-return condition is what the shift-size sweep below is finding numerically at its peak (and what pymc-marketing’s BudgetOptimizer solves directly in production).

TV ROAS - average 2.08x * marginal (the next euro, at today's spend) 0.23x - marginal is lower because of saturation (diminishing returns).

Shifting 15% of brand_search spend (€566) to TV -> +€61 net sales (saturation-aware); the sweep peaks near 22%.

$P(\text{TV ROAS} > \text{brand_search ROAS}) = 0.78$ -> lean toward TV, but test first

Ranking vs levels - the watertight part:

TRUE ROAS: TV 0.85x * brand_search 0.14x (both BELOW 1x) | MMM says TV 2.08x * bs 0.60x

The MMM recovers the RANKING (TV >> brand_search) but overstates LEVELS ~2.4x for TV.

ROAS is revenue / spend - it assumes 100% margin. At a realistic 30% margin the break-even is $1/0.30 \sim 3.3x$,

which even TV's estimated 2.08x fails; and TV's TRUE ROAS 0.85x is below break-even at ANY margin.

So: reallocate toward TV (the ranking is trustworthy), but 'is TV worth it in absolute €?' needs the geo

experiment (nb 07) - this MMM's absolute levels are not calibrated enough to answer it.

Decision quantity WITH uncertainty - shift 15% (€566)

brand_search -> TV (per-draw, average-ROAS):

$\text{Deltasales} = €836$ [90% €-799, €4,618] * $P(\Delta > 0) = 0.78$

TRUE Deltasales (from DGP true_contrib) = €405 -> direction check: MATCH (both favour TV)

THE TWO Deltasales FIGURES cell 32 weighs, same 15% shift:

- saturation-aware (walks both channels DOWN their fitted curves) : €61/week
<- the honest planning number

- average-ROAS (ignores saturation, carries the uncertainty band):
€836/week [90% €-799, €4,618], $P(\Delta > 0) = 0.78$

Pricing the experiment (the notebook's punchline, in euros): acting on

the DEFAULT MMM means shifting 15% of TV spend (€926) into

brand_search -> Deltasales = €-662 per planning window (value DESTROYED);

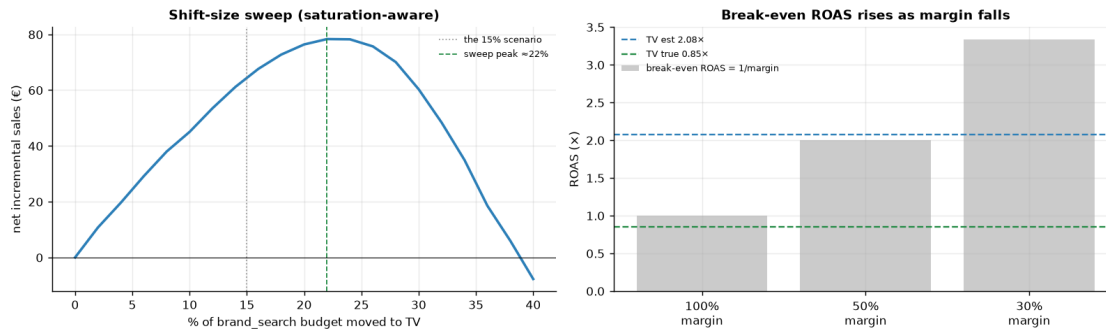
acting on the CALIBRATED MMM adds €405. The swing, €1,067

per window - recurring every window - is a FLOOR on what the geo experiment

that calibrated the prior was worth (it prices only this one 15%

decision, nothing else the calibration de-risks). If the test costs less

than a few windows of that, it pays for itself.



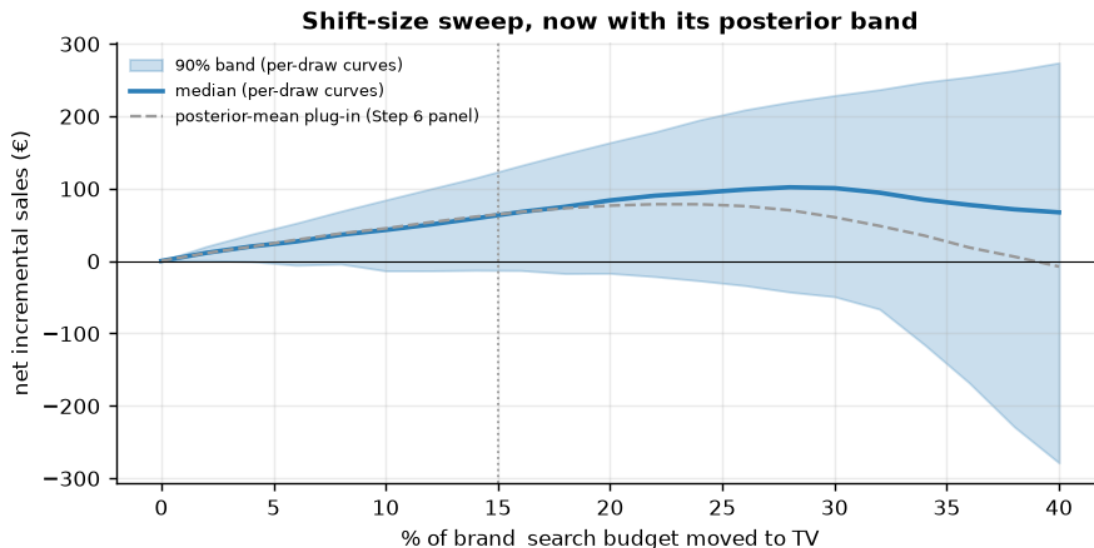
Which Δ sales number goes to the CFO? The block just above prints two Δ sales figures for the same 15% shift, and they differ by roughly an order of magnitude *on purpose*. The **saturation-aware** figure walks both channels *down* their fitted response curves — at TV’s current operating point the marginal curve is already nearly flat, so the honest planning number is small (in the committed FULL run $\approx +\text{€}61/\text{week}$; a FAST run prints an even smaller, noisier value that can dip slightly negative). The **average-ROAS** figure prices the same shift at each posterior draw’s *average* ROAS — it ignores saturation entirely, so it lands far larger ($\approx +\text{€}836/\text{week}$ in the FULL run, $\sim 14\times$ the saturation-aware figure), and is kept only because it carries the uncertainty band: its printed $P(\Delta > 0)$ is the **ranking probability in euro clothing** — by construction $\Delta > 0$ exactly when TV’s ROAS beats brand_search’s. **Quote the saturation-aware figure to the CFO; use the average-ROAS band as the direction check it is.**

Giving the planning number its band

One loose end, flagged honestly: the saturation-aware sweep above was built by re-fitting a Michaelis–Menten curve to the **posterior-mean** weekly contributions with `scipy.curve_fit` — a plug-in convenience that throws the posterior away, which is why the saturation-aware planning number above printed with no interval while the cruder average-ROAS number carried one. A notebook that preaches posteriors should not quietly switch to point estimates for its decision-critical figure. The fix costs almost nothing: repeat the same curve fit on **each posterior draw** of the calibrated MMM’s weekly contributions, push every draw through the same budget shift, and the sweep inherits a genuine 90% credible band.

Saturation-aware Deltasales at the 15% shift: median €59 [90% €-13, €115] *
 $P(\Delta > 0) = 0.90$ (from 300 of 300 draws)

This is the CFO sentence completed: quote the saturation-aware median WITH its band, alongside the ranking probability - the band separates conviction from hope.



The one-paragraph decision

Recommendation: act on the ranking, pilot the shift, and let a geo test set the levels. The calibrated MMM’s channel **ranking** — TV above brand_search — is the trustworthy part of this analysis: it is direction-checked against the planted truth, it survives the entire experiment-plausible prior range in View 6, and the posterior predictive check showed the competing default-prior reading buys no extra fit — the experiment, not the data, is what identified it. *How much* to move is where honesty bites: judge the printed $P(\text{TV beats brand_search})$ against the 0.8 action bar (in the committed FULL run it lands just short, at ≈ 0.78 — hence “pilot”, not “reallocate”), and size the pilot from the **saturation-aware sweep with its band**, never from average ROAS — in the FULL run that sweep peaks at a roughly one-fifth shift, while a FAST run’s short chains are ranking-grade only and should not size anything. The absolute euro **levels** are explicitly not decision-grade: the MMM overstates contributions several-fold even after calibration, and the *true* TV ROAS sits below break-even at any margin, so “is TV worth it at all?” belongs to the geo experiment (nb 07) — whose per-window value the swing computation above already floors in euros, recurring every planning cycle. Act on the ranking; pilot the shift; let the experiment set the levels.

```
{
  "true_contribution_eur": {
    "tv": 5263,
    "brand_search": 518
  },
  "mmm_default": {
    "tv": 10390,
    "brand_search": 36007,
    "ranking_correct": false,
    "convergence": "max r-hat 1.010 - min ESS 381 - divergences 14"
```

```

},
"mmm_calibrated": {
  "tv": 12820,
  "brand_search": 2262,
  "ranking_correct": true,
  "convergence": "max r-hat 1.010 - min ESS 376 - divergences 0"
},
"ppc_share_in_90pct_band": {
  "default": 0.89,
  "calibrated": 0.9
},
"p_rank_correct_calibrated": 0.78,
"action_bar": 0.8,
"prior_sweep": {
  "grid_mu_bs": [
    0.15,
    0.3,
    0.6,
    1.0,
    2.0
  ],
  "p_tv_beats_bs": [
    0.93,
    0.86,
    0.39,
    0.21,
    0.12
  ],
  "flips_at_mu": 0.6
},
"chosen_shift_of_bs_budget": 0.15,
"delta_sales_saturation_aware_eur": {
  "median": 59,
  "lo90": -13,
  "hi90": 115,
  "p_positive": 0.9
},
"delta_sales_avg_roas_eur": {
  "mean": 836,
  "lo90": -799,
  "hi90": 4618
},
"true_delta_sales_eur": 405,
"experiment_value_floor_eur_per_window": 1067,
"verdict": "pilot shift toward TV (ranking trustworthy); geo test for absolute
levels"
}

```

7 · Caveats

- **Absolute contributions need calibration.** Even the calibrated MMM runs a few times too high in absolute euros, because channel spend rides the seasonal wave the model can't fully net out from a short series. **Trust the ranking and the ROI verdict, not the absolute euros** — and note it took an experiment-informed prior to get even the *ranking* right. This is the single most important thing to say about any MMM.
- **A DAG is necessary but not sufficient.** The correct backdoor control (seasonality) did not stop a flexibly-saturated, collinear channel from re-absorbing the confounder. Structural assumptions still need experimental calibration to be trustworthy.
- **Fixing the transform does not remove its uncertainty — it hides it.** Step 0's classical MMM grid-searches λ and κ and then reports OLS intervals *conditional* on them. Its profile-likelihood region showed those constants are barely identified at all, so that interval understates total uncertainty by a large factor (printed there) and can exclude the truth outright. If you inherit an MMM whose adstock and saturation were “set to industry standards”, the intervals on its ROAS are decorative.
- **Adstock/saturation priors matter — a lot.** Carry-over (λ) and diminishing-returns (α , κ) are weakly identified from limited weekly data; here the saturation-ceiling prior *flipped the conclusion*. Use informative priors from experiments wherever you have them.
- **Correlated channels → wide, correlated ROI posteriors.** When spend moves with a confounder, the data can't fully separate them; the intervals will (honestly) show it.